

MX-TreeFI: Value-Aware Statistical Fault Injection for Microscaling (MXFP) Neural Networks

Problem Statement, Methodology & Fault-Rate Model — Working Draft

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Abstract

Reliability evaluation of deep neural networks under hardware faults relies on fault injection (FI), but exhaustive campaigns are intractable and statistical fault injection (SFI) wastes budget by ignoring that a bit flip's effect depends on the value being corrupted. TreeFI exploited this value-dependence for FP32, partitioning each layer's value distribution into intervals with similar bit-flip behavior and allocating injections by frequency and uncertainty, cutting the required campaign by up to 72×. However, TreeFI assumes each value is a self-contained 32-bit word flipped by a strict, position-uniform single bit — assumptions that break in **MXFP microscaling block floating-point**, where a block of $K = 32$ elements shares one 8-bit scale and each element is only 4–8 bits. We propose **MX-TreeFI**, a value-aware statistical fault-injection methodology for MXFP that treats the fault space as two coupled strata — the per-block shared scale and the per-element value — and allocates a shared injection budget across both. We (i) redefine the fault model with a per-bit-position flip rate and a multi-bit extension (the H-model), (ii) generalize TreeFI's value-aware interval learning to the joint (scale, element) decoded value, and (iii) extend the stratified allocation with blast-radius and input-sensitivity weighting. We formalize the occurrence model via a Poisson–binomial multiplicity expansion and integrate it with TreeFI's uncertainty-based allocation, and outline an evaluation across MXFP8/6/4 on CNN and transformer models against exhaustive FI and SFI/IFI baselines.

Keywords — statistical fault injection, MXFP microscaling, block floating-point, DNN reliability, value-aware, bit-flip fault model, soft errors.

1 Problem statement

TreeFI rests on two assumptions that hold in FP32 but break in MXFP: that each value is a **self-contained 32-bit word**, and that faults are **strictly single-bit with every bit position equally likely**. MXFP — now standard for efficient Vision Transformers, LLMs, CNNs, and GNNs — violates both. A block of $K = 32$ elements shares one 8-bit scale R_s , each element is a narrow **4/3/2-bit value** $r_{i,s}$ (the (8,4), (8,3), (8,2) configs), and the decoded value composes both levels: $v_{i,s} = f(R_s, r_{i,s})$.

ASSUMPTION	FP32 — TREEFI	MXFP — MX-TREEFI
Value word	Self-contained 1+8+23 bits	Two levels: shared scale R + narrow element r
Fault sites	One (the value itself)	Two coupled sites — element and shared scale
Blast radius	1 value per flip	1 value (element) vs 32 values (scale)
Bit occurrence	Uniform across positions	Per-bit-position flip rate π^b
Value dependence	Per-scalar value	Joint over $(R_s, r_{i,s})$
Input dependence	Per-input activations	Inputs/blocks weighted by relative sensitivity

MX-TreeFI models the MXFP fault space as two coupled strata — per-block shared scale and per-element value — weights fault occurrence by a per-bit-position flip-rate model, and allocates a shared injection budget across sites, bit positions, and value ranges (by value frequency, estimation uncertainty, blast radius, and relative input sensitivity) to estimate the failure rate at a target confidence with far fewer injections than uniform SFI.

Research questions

- RQ1 Fault model.** Define an MXFP fault model combining two sites (scale R / element r), a per-bit-position flip rate π_b per site, and single vs multi-bit flips (*open* — *H-model*).
- RQ2 Value-awareness.** How does joint $(R_s, r_{i,s})$ dependence shape impact — and should intervals be *learned* (regression tree) or *enumerated exactly* given 2–4-bit elements? (*open*)
- RQ3 Budget allocation.** How to split injections across sites, bit positions, and value ranges — extending $n \propto W \cdot \sqrt{(r(1-r))}$ with π_b , blast radius, and input sensitivity.
- RQ4 Quality vs cost.** Does it hold TreeFI-level quality ($\leq 1\%$ MAE at target confidence) while matching or exceeding its injection reductions across MXFP8/6/4, activations and weights?

2 Methodology

TreeFI's two-phase structure is preserved — an offline characterization pass followed by a stratified injection campaign — but every step generalizes to the two MXFP sites. The guiding separation: an **occurrence model** (π_b — where and how often flips happen) multiplied by an **impact model** (value-aware severity — how much they hurt).

Phase I – Offline characterization

(a) Decode & collect distributions. Per target layer, accumulate the element value/code histogram and the shared-scale R histogram. Both alphabets are tiny (scale ≤ 256 ; element 4–16 codes for MXFP4/6/8), so characterization can be near-exact rather than sub-sampled.

(b) Fault model — two sites, per-position rates. Single-bit flips at two sites, each with a per-position rate $\pi_{\text{site},b}$: an **element flip** (bit b of $r_{i,s}$) has local severity $d = |\log v'_{i,s} - \log v_{i,s}|$; a **scale flip** (bit b of R_s) rescales every element by $2^{\Delta R}$, giving block severity $D_s = \sum_i |\log v' - \log v|$ — quantifying the $\sim 32\times$ blast radius. Open (RQ1c): a multi-bit “H-model” layered on the single-bit base (§3).

(c) Impact characterization (RQ2, open). Compare two abstractions: TreeFI-style **learned value intervals** (regression tree over decoded values) vs **exact per-code enumeration** over $(R_s, r_{i,s}, b)$ — feasible given the tiny element alphabets.

(d) Risk scoring with blast radius. Extend $r = \sigma(s \cdot (\text{avg_log} - \beta))$: element intervals scored on local severity; scale codes scored on block-aggregated severity D_s , so scale bits carry structurally higher risk — placed on a common scale with element risk.

Phase II – Stratified injection campaign

(e) Budget allocation (RQ3). Partition the population by site $\rightarrow$ bit position $\rightarrow$ value interval, and allocate:

$$\eta_{L,site,b,k} \propto \pi_{site,b} \cdot W_{L,site,k} \cdot \sqrt{r_{L,site,b,k}(1-r_{L,site,b,k})} \cdot \Phi(\text{blast}_{site})$$

π = per-bit-position flip rate · W = value-interval frequency · $\sqrt{r(1-r)}$ = TreeFI uncertainty term · Φ = blast-radius up-weight for the scale stratum. Total fixed by target confidence / error.

(f) **Execution. Element faults:** pick locations whose decoded value falls in the target interval (activations per input; weights per parameter), flip the chosen bit. **Scale faults:** pick a block whose scale falls in the target interval, flip a scale bit — perturbing all 32 elements at once.

(g) **Aggregation with input sensitivity.** Recombine per-interval failure rates by frequency within each stratum, combine the two strata, and weight inputs/blocks by relative sensitivity ϕ_j (the $I_4 : 4I_1 = 3I_2 = 2I_3$ observation):

$$\hat{p}_L = \sum_j \phi_j \cdot \sum_{site} \sum_b \sum_k \pi_{site,b} \cdot W_{L,site,k} \cdot \hat{p}_{L,site,b,k}$$

The estimate reflects value frequency, per-position occurrence, site population, and that some inputs/blocks contribute more to the failure rate than others.

3 Fault-rate model, formalized

Step (b) named two ingredients — a per-position flip rate π_b and an open multi-bit “H-model.” Here both are made precise. Throughout, **occurrence** (where and how often a flip lands — a hardware, position property) is kept strictly separate from **impact** (how much it hurts — the value-aware risk r of §2d). This section derives occurrence only; the two multiply in the campaign.

3.1 Per-position flip rate

Each site — scale R (width $W_R = 8$) and element r (width $W_r = n \in \{4,3,2\}$) — assigns every bit position an independent flip probability, factored into a site base rate and a positional profile:

$$\pi_{site,b} = \pi_{site}^0 \cdot \kappa_{site}(b), \quad \text{with} \quad (1/W) \sum_b \kappa_{site}(b) = 1$$

π_{site}^0 = site base per-bit rate · $\kappa(b)$ = positional susceptibility profile (mean 1). A graded κ ascending across positions reproduces the notes' 1% / 2% / 3% / 4% ladder; $\kappa \equiv 1$ recovers the classic uniform single-bit assumption.

3.2 Word multiplicity — Poisson–binomial

Bits flip independently, so a word's outcome is a flip pattern S over its bit positions. This is the machinery behind the notes' single- and two-bit ladders:

$$P(S) = \prod_{b \in S} \pi_b \cdot \prod_{b \notin S} (1 - \pi_b) \quad P_m = \sum_{|S|=m} P(S)$$

$P(S)$ = probability of exactly that flip pattern · P_m = probability of exactly m simultaneous flips.

- **Single-bit** ($m=1$, TreeFI's regime): $P(b \mid m=1) = \pi_b \prod_{b' \neq b} (1 - \pi_{b'}) \approx \pi_b / \sum_{b'} \pi_{b'}$.
- **Two-bit** ($m=2$): $P_2 \approx \sum_{b < b'} \pi_b \pi_{b'}$; for equal rates $\rightarrow C(W,2) \pi^2$. So $(1\%)^2 = 0.01\%$ — exactly the notes' 2-bit ladder (0.01–0.04%).

3.3 The H-model (word-corruption rate)

The aggregate probability that a word is corrupted at all, expanded by multiplicity for small rates:

$$H_{\text{site}} = 1 - \prod_b (1 - \pi_b) = P_1 + P_2 + \dots \\ \approx \sum_b \pi_b + \sum_{b < b'} \pi_b \pi_{b'} + O(\pi^3)$$

First term = single-bit exposure; second = multi-bit tail. Instantiating the dominant positions at $\approx 1\%$ and letting the tail scale with word width M gives the notes' closed form:

$$H(M) \approx 3 \cdot (1\%) + 0.01\% \cdot M$$

$3 \cdot 1\%$ = dominant single-bit exposure ($\approx \sum \pi_b$ over the significant positions) $\cdot 0.01\% \cdot M$ = multi-bit tail growing with word width M .
Single-bit regime: drop the second term, $H \approx \sum_b \pi_b$.

3.4 Block coupling & blast radius

An MXFP block is 8 scale bits + 32·n element bits, so scale and element exposure enter the same block at very different weights once blast radius is applied:

$$E[\text{flips}]_{\text{block}} = H_R + 32 \cdot H_r \rightarrow E[\text{impact}]_{\text{block}} \propto K \cdot H_R + 32 \cdot (1) \cdot H_r = 32 \cdot (H_R + H_r)$$

Raw flip counts favor the 32 element words by $\sim 32\times$. But each scale flip hits $K = 32$ values (blast K) and each element flip hits 1, so impact-weighting the single shared-scale word by K restores parity between the two sites. This K factor is exactly what $\Phi(\text{blast})$ carries in §2e.

3.5 Into the campaign

The formalized occurrence mass now replaces the placeholder $\pi_{\text{site},b}$. In the general multi-bit form the unit is a pattern S ; the single-bit regime collapses it back to §2e:

$$n_{L,\text{site},S,k} \propto P(S) \cdot W_{L,\text{site},k} \cdot \sqrt{r(1-r)} \cdot \Phi(\text{blast}(S, \text{site}))$$

Single-bit regime $S = \{b\}$: $P(S) \approx \pi_{\text{site},b}$, recovering the earlier allocation exactly. Aggregation then weights each measured rate $\hat{p}$ by its occurrence probability $P(S)$ rather than a flat injection count.

4 Evaluation plan

FORMATS	MXFP8 (E4M3 / E5M2), MXFP6 (E3M2 / E2M3), MXFP4 (E2M1) — the (8,4) / (8,3) / (8,2) configs; E8M0 scale, block 32.
DOMAINS	Activations and weights, both fault sites.
GROUND TRUTH	ResNet8 (CIFAR-10) exhaustive reference — feasible given MXFP's small bit space — scaling to DeiT / a small LLM where exhaustive FI is not.
BASELINES	MXFP-adapted uniform SFI and iterative IFI.
METRICS	Injection-count reduction at fixed confidence / error; MAE vs exhaustive; per-site, per-bit, per-layer fidelity.
ABLATIONS	Blast-radius weight Φ , per-position rates π_b , and single- vs multi-bit — mirroring TreeFI's hyperparameter study.

Deliberately open, to be settled empirically as ablation studies: **RQ1c** — single-bit base vs multi-bit “H-model” fault occurrence; **RQ2** — learned regression-tree intervals vs exact per-code enumeration for the narrow element formats.